

# V.A.N.I - xAI

## VĀṆĪ ADHYĀTMĪK NAVĪN INTELLECT

A Flagship Sovereign Innovation under the **Bureau Of V.A.N.I-xAI (BoVxAi)** • Next-Gen 100% Free AI Tutoring, Multimodal Socratic Mentorship & Autonomous Learning for Bharat.



### Innovator & Institution

- **Innovator:** Dhruv Sagar (Class 10th)
- **Institution:** PM SHRI KV No. 1 AFS Chakeri, Kanpur (UP)
- **Student UID:** 1001049893

### Core Innovation Pillars

- **100% Free Sovereign Access:** Zero subscriptions, zero paywalls.
- **Asymmetric Sub-User Architecture:** <50KB client load on 2G/3G.
- **Socratic Mentorship:** Guided concept scaffolding, not a cheat engine.
- **100% Zero E-Waste:** Software-driven sustainability model.

# Innovator & Bureau Of V.A.N.I-xAI (BoVxAi)

Grassroots technological leadership from PM SHRI KV No. 1 AFS Chakeri, Kanpur



## The Student Innovator

### Dhruv Sagor

Class 10th student with deep specialization in Applied Machine Learning, Full-Stack Architecture, and Asymmetric Cloud Systems. Built V.A.N.I-xAI from scratch to solve educational inequality.



## PM SHRI KV No. 1 Chakeri

Selected under the Government of India's prestigious **PM SHRI Scheme** for exemplary educational standards. Fosters student research in science, robotics, and societal technology.



## Bureau Of V.A.N.I-xAI

**BoVxAi** is the parent innovation cell dedicated to engineering sovereign, accessible AI algorithms. VANI AI serves as its primary educational deployment for Indian classrooms.

# The Digital & Pedagogical Divide in India

Why commercial AI platforms fail 250M+ grassroots school students across Bharat



## 1. Severe Paywalls & Exclusivity

Commercial AI tools (ChatGPT Plus, Claude Pro) mandate recurring subscriptions of ₹1,500 – ₹2,500/month and require international credit cards. This locks out millions of talented rural students.



## 2. Hardware & Bandwidth Barriers

Modern AI applications require heavy GPU RAM, modern flagship smartphones, and high-speed broadband. Underprivileged students using ₹5,000 budget phones and 2G/3G suffer crashes.



## 3. The "Cheat-Engine" Pedagogical Flaw

Mainstream LLMs generate direct homework answers instantly. This encourages copy-pasting and cognitive laziness rather than critical thinking, problem-solving, and concept mastery.



## 4. Lack of Vernacular & Safe Guardrails

Foreign AI platforms lack alignment with Indian NCERT/CBSE curricula, vernacular voice support for Hindi/regional dialects, and safety guardrails tailored for young school children.

# V.A.N.I-xAI: Sovereign AI Mentor for Bharat

Democratizing cloud intelligence through indigenous asymmetric architecture



## <50KB Client Core

99% of compute is offloaded to cloud servers.  
Web client loads in sub-seconds even on legacy  
₹5,000 mobile phones and slow 2G/3G  
connections.



## Socratic Mentorship

Never gives direct answers. Instead, it guides  
students step-by-step with interactive hints,  
Socratic questions, and concept breakdowns.



## BoV Multimodal Voice

Integrated voice synthesis (`voice_agent.py`)  
enabling real-time auditory tutoring for  
auditory learners and visually challenged  
students.



## 100% Free Sovereign

Completely eliminated all paywalls, premium  
plans, and credit card friction. Permanently  
free for all Indian students and teachers.



# Bureau Of V.A.N.I-xAI (BoVxAi) Novel IP

Handcrafted algorithmic systems engineered from scratch for the V.A.N.I ecosystem

## 1. Mathematical UID

Algorithmically generates permanent tokens (e.g. V.A.N.I-xAI-DHRU-3478) binding persistent memory and multi-device sessions without hardware lock-in.

## 2. 3-Layer Security

Proprietary algorithmic authorization validated against: (1) DB check, (2) Active student token verification, and (3) UID sandbox isolation.

## 3. InAixVAi Dashboard

A bespoke educator/admin dashboard granting real-time classroom telemetry, query load oversight, token analytics, and sandbox security controls.

## 4. Autonomous Memory

Self-healing background engine managing cross-device sync, SQLite memory (`vani_memory.db`), and server load balancing with zero human maintenance.



# Multi-Agent Intelligence & Backend Modules

State-of-the-art intelligent engines integrated in the local & cloud backend



## LLM Router & SQLite Memory (`llm_router.py`)

Dynamic multi-model fallback across vision and text models. Powered by `vanila_memory.db` for persistent contextual retention and lightning-fast semantic search.



## Multimodal Voice Hub (`voice_agent.py`)

Integrates edge TTS to convert scientific derivations, diagrams, and guided explanations into crystal-clear speech for inclusive hands-free learning.



## Vision-Guided Agentic Loop (`agentic_loop.py`)

Performs visual reasoning on user screen states, formulas, and diagrams, converting V.A.N.I into an interactive digital lab assistant.



## Proactive System Agent (`autonomous_agent.py`)

Monitors system battery status, initiates eco-friendly power optimization, and schedules automated learning reminders for students.



# 5-Step Zero-Friction Operational Pipeline

Instant onboarding with mathematical security and lifelong learning sync

## STEP 01

### Auth & UID Gen

Student signs in securely via Google OAuth. System computes and assigns permanent UID.

## STEP 02

### Instant Launch

Launches workspace with 100% unlocked features, zero fees, zero subscription friction.

## STEP 03

### Swarm Sync

Binds UID to encrypted session tokens, enabling cross-device sync & persistent state.

## STEP 04

### Socratic Tutoring

Queries route through LLM router, memory DB, and Socratic scaffolding filters.

## STEP 05

### Lifelong Memory

Autonomous memory engine records learning milestones for continuous concept mastery.

# V.A.N.I-xAI vs Conventional Platforms

Engineered specifically for maximum grassroots accessibility and pedagogical integrity

| Evaluation Metric         | Commercial AI (ChatGPT/Claude)    | Traditional EdTech Portals        | V.A.N.I - xAI (BoVxAI Innovation)       |
|---------------------------|-----------------------------------|-----------------------------------|-----------------------------------------|
| Cost for Students         | ₹1,500 – ₹2,500 / month           | ₹5,000 – ₹25,000 / year           | 100% Free Lifetime (Zero Paywalls)      |
| Hardware / Device Barrier | Requires modern smartphone RAM    | Heavy 100MB+ app download         | Runs in any browser (<50KB payload)     |
| Pedagogical Approach      | Direct answer output (cheat risk) | Static pre-recorded videos        | Socratic Guidance & Concept Scaffolding |
| Payment & Credit Barrier  | Mandates International Cards      | Closed recurring subscriptions    | Zero Paywalls / Instant Free Access     |
| Zero E-Waste & Ecology    | Heavy local battery & GPU drain   | Proprietary locked tablet devices | 100% Cloud-Native (Zero E-Waste)        |



# 100% Zero E-Waste & Eco-Friendly Architecture

A sustainable software innovation aligned with National Green Missions



## Zero Physical Manufacturing

Requires zero plastic injection molding, toxic lithium batteries, or PCB scrap. Completely eliminates electronic waste at the manufacturing root.



## Extends Old Hardware Lifespan

Breathes new life into existing school computer labs and parents' low-cost old smartphones, preventing premature hardware disposal.



## 100% Paperless Ecosystem

Eliminates physical question banks, printed worksheets, and test guides through dynamic cloud mentoring and digital session memory.



# Grassroots Value & ₹10,000 DBT Fund Budget

Resource Optimization & Grassroots Scaling Plan for Bharat



## Grassroots Societal Value

- **Democratic Access:** Delivers 24/7 personalized 1-on-1 STEM mentoring to Tier-2, Tier-3 & rural learners with ₹0 fees.
- **Empowers First-Gen Learners:** Deep conceptual clarity in Science, Math & Coding.
- **NEP 2020 Alignment:** Promotes inquiry-based, experiential learning over rote memorization.



## DST ₹10,000 DBT Fund Plan

- **₹5,000:** Cloud API credits (empowers 100,000+ student educational queries).
- **₹2,500:** Serverless database hosting & low-latency Indian Edge CDN scaling.
- **₹1,500:** Vernacular token optimization for Hindi & regional Indian dialects.
- **₹1,000:** SSL security, domain infrastructure & school cluster deployment.

# Working Prototype & Validation Status

Tested, verified codebase ready for school & district implementation



## Live InAixVAi Panel

Active teacher command console tracking user queries, active sessions, and security isolation.



## SQLite Memory Engine

vani\_memory.db operational with sub-5ms query retrieval and zero context loss.



## Edge TTS Voice Agent

Real-time speech synthesis enabled for conversational bilingual explanations.



## Cross-Device Ready

Verified across low-cost Android, Windows PC, Raspberry Pi, and iOS browsers.



# Vision 2030: From School Lab to National Utility

Structured scale-up strategy for NCSC & National Education Missions

## Phase 1: School Pilot (2026)

- Sovereign sub-user engine deployed
- Socratic AI engine with zero-cost access
- Pilot launch at PM SHRI KV No. 1 Chakeri

## Phase 2: District & State

- 10+ Vernacular Indian languages
- Bi-directional voice conversation
- State-wide KV & Govt school rollout

## Phase 3: National Scale

- Integration with DIKSHA & PM eVidya
- Interactive AI virtual lab assistant
- Atmanirbhar Bharat AI mentoring stack

## "Empowering Every Student with Sovereign AI Mentorship"

Thank you, Respected Jury Members & DST / NIF Mentors • Bureau Of V.A.N.I-xAI (BoVxAI)